Exercise Sheet 5

1. For an MA(1) process $X_t = Z_t + \theta Z_{t-1}$ we obtained by applying the innovations algorithm that

$$\hat{X}_{n+1} = \frac{\sigma^2\theta}{\nu_n}(X_n - \hat{X}_n),$$

with

$$\nu_n = \sigma^2\left(1 + \theta^2 - \frac{1}{\nu_{n-1}}\theta^2\sigma^2\right),$$

and $\nu_0 = \sigma^2(1+\theta^2)$. Show that if $|\theta| < 1$ then $|\nu_n - \sigma^2| \leq C\rho^n$ for some $0 < \rho < 1$ and some large enough constant C (exponentially fast convergence). Also the convergence $\theta_{n1} \to \theta$ is exponentially fast.

What happens to ν_n if $|\theta| > 1$ or if $|\theta| = 1$?

2. Consider observations $X_1, \ldots, X_n$ of the causal and statioanry AR(p) process

$$X_t - \phi_1 X_{t-1} - \cdots - \phi_p X_{t-p} = Z_t, \quad \{Z_t\} \overset{iid}{\sim} N(0,1).$$

Show that the likelihood function is given as

$$L(\phi_1, \ldots, \phi_p | X_1, \ldots, X_n) = (2\pi)^{-n/2}|\Gamma_p|^{-1/2} \times \exp\left(-\frac{1}{2}\left[\mathbf{X}_p^T\Gamma_p^{-1}\mathbf{X}_p + \sum_{t=p+1}^{n}(X_t - \phi_1 X_{t-1} - \cdots - \phi_p X_{t-p})^2\right]\right),$$

where $\mathbf{X}_p = (X_1, \ldots, X_p)'$ and Γ_p is defined as usual. Conclude that

$$-\frac{2}{n}\log L(\phi_1, \ldots, \phi_p | X_1, \ldots, X_n) - \frac{1}{n}\sum_{t=p+1}^{n}(X_t - \phi_1 X_{t-1} - \cdots - \phi_p X_{t-p})^2 \to 0.$$

Interpret this latter result.

3. Consider the time series $X_t = Z_t + \theta Z_{t-1}$ where $\{Z_t\} \sim \mathrm{WN}(\sigma^2)$. Assuming we have an invertible model, derive moment estimators for θ and σ^2.